

Quiz 9

● Graded

Student

Samyak Sanghvi

Total Points

3 / 4 pts

Question 1

Q1

1 / 2 pts

✓ + 1 pt Without Gaussian smoothing: spurious edges detected due to noise

+ 1 pt Without non-max suppression: detected edges are thick instead of 1 pixel wide

+ 0 pts No/Wrong answer

Question 2

Q2

2 / 2 pts

+ 0 pts No/Wrong answer

Either

✓ + 1 pt $\sigma_B^2 = P1(k) P2(k) (m1(k) - m2(k))^2$

✓ + 1 pt if $k \rightarrow 0$ or 255 then $P1 \rightarrow 0$ or $P2 \rightarrow 0$ respectively

Or

+ 1 pt $\sigma_B^2 = P1 (m1 - mG)^2 + P2 (m2 - mG)^2$

+ 1 pt if $k \rightarrow 0$ then $P1 \rightarrow 0$ and $m2 \rightarrow mG$ and vice versa when $k \rightarrow 255$

Name: Samyukh Sanghvi

COL7(6)83 Quiz 9

Entry no.: 2023(S1080)

31 October, 2025

1. What would be the downside if Canny edge detection were performed without Gaussian smoothing? What about without non-maximum suppression?

If we were to do canny edge detection w/o smoothing the noise or some unnecessary features still remain in the image ruining the differences and creating ~~pr~~ unnecessary phantom edges or cause ~~the~~ unusually large gaps.

2. In the context of intensity thresholding, explain why the between-class variance $\sigma_B^2(k)$ decreases towards zero towards both ends of the intensity range (i.e. as k approaches 0 or 255 for an 8-bit image).

$$\sigma_B^2 = P_1 P_2 (m_1 - m_2)^2$$

as we go to 0 or 255

$P_1, P_2 \rightarrow 0$ respectively.

$$\therefore \sigma_B^2 \rightarrow 0$$